

Lab 3 – Municipal Infrastructure Damage Reporting App

Architecture Selection

We chose Layered Architecture for the Municipal Infrastructure Damage Reporting App.

Architectural Choice

The system is divided into a Presentation Layer, Business/Service Layer, and Data/Integration Layer. The Presentation Layer contains the Resident/Staff Interface. The Business/Service Layer contains the Damage Report Manager. The Data/Integration Layer contains the Location & Media Service, Notification Service, and Municipal Report Database. The components communicate through clearly defined provided and required interfaces.

Reason 1 – Separation of Responsibilities

Layered architecture separates user interaction, business logic, and persistent data. The interface collects damage reports, the Damage Report Manager validates and manages them, and the database stores report information. Location/media and notification functionality is kept behind separate service interfaces.

Reason 2 – Suitable for the Reporting Workflow

The reporting process is structured: a user submits damage information, the application manages the report, supporting services handle location/media and notifications, and the report is stored. Layered architecture maps directly to these responsibilities and keeps the design easy to understand and maintain.

Security Advantage

The Damage Report Manager acts as the controlled application boundary for report data. Clients do not need direct access to the database; access can be restricted through authentication and authorization at the application layer, including different permissions for residents and municipal staff.

Performance Benefit

The main submission and status workflow is kept within a small number of focused components. The application can request database, location/media, or notification services only when required, avoiding unnecessary direct communication between clients and backend resources.